

1. What is the 4-bit output of adding the following unsigned binary integers?
 - a. 1111 and 1110
 - b. 1001 and 1111
2. Write the two's complement representation of the following eight-bit values:
 - a. -10
 - b. -12

1. a) 1=1, 2=0 carry 1, 3=1 carry 1, 4= 1 carry 1, 5 = 1
11101
- b) 1=0 carry 1, 2=0 carry 1, 3=0 carry 1, 4=1 carry 1, 5=1
11000
2. a) $10 \% 2 = 5,0$
 $5 \% 2 = 2,1$
 $2 \% 2 = 1,0$
 $1 \% 2 = 0,1$
 1010
 00001010
 11110101
 00000001
 11110110
- b) $12 \% 2 = 6,0$
 $6 \% 2 = 3,0$
 $3 \% 2 = 1,1$
 $1 \% 2 = 0,1$
 1100
 00001100
 11110011
 00000001
 11110100